

Image-Based Data Loss Prevention: Combining OCR-NLP Analysis with Deep Learning Steganalysis

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Abstract :

Enterprise Data Loss Prevention (DLP) for unstructured digital images faces two main types of threats: the obvious leaking of sensitive visual or textual information and hidden data theft using digital steganography. Traditional tools that rely on pattern matching and standalone Optical Character Recognition (OCR) struggle significantly when dealing with complex layouts, handwritten text, or hidden data within images. This research introduces a combined DLP system that addresses these challenges through multi-modal analysis and deep learning techniques. The first part uses OCR (Tesseract) combined with pattern-based detection, contextual keyword verification, and Named Entity Recognition (NER) to identify sensitive information in visible text, supported by a lightweight OpenCV structural classifier (optionally enhanced by CLIP ViT-B/32 zero-shot classification) for document-type estimation and an adaptive preprocessing pipeline to improve OCR accuracy. The second part employs an EfficientNetV2-S convolutional network adjusted to recognize covert data hiding methods in both spatial and frequency domains. Tests across various document types (like bank statements, police reports, medical prescriptions, and ID cards) and multiple steganographic datasets show the system achieves 78.40% validation accuracy on five-class steganalysis while effectively identifying obvious privacy breaches.

Keywords : Data Loss Prevention (DLP), Steganalysis, Multi-Modal Learning, Document Classification, Deep Learning, Image Forensics, Zero-Shot Inference.

I. INTRODUCTION

Businesses in the digital world, healthcare networks, and financial institutions deal with large amounts of unstructured image records such as scanned legal documents, handwritten prescriptions, banking statements, and identification cards. Laws like the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) require strict protection of Personally Identifiable Information (PII) and Protected Health Information (PHI). However, using unstructured visual formats creates significant risks for unauthorized data leaks and stealthy data theft. Current DLP methods mainly depend on string searches, regular expressions (Regex), and standard OCR text extraction. While these approaches work well for standard

digital documents encoded in ASCII/Unicode format, they falter when faced with unstructured visuals that have skewed layouts or varying fonts. Moreover, traditional text-focused DLP wrongly assumes all stolen data is clearly visible. Malicious actors find ways to bypass security by hiding sensitive information in pixel arrangements or frequency coefficients of ordinary-looking images using steganographic techniques like LSB, PVD, DCT, and FFT. Standard inspection tools are unable to perform detailed pixel-level audits; therefore, covert channels often evade detection.

To tackle these dual issues, this paper proposes a thorough defense architecture with multiple layers. The key contributions are:

1. An OpenCV-based structural document-type classifier, optionally enhanced by CLIP (ViT-B/32) zero-shot classification, that estimates document type (e.g., ID card, passport, cheque, screenshot) as supporting evidence alongside OCR-based analysis.
2. An adaptive image preprocessing system combining Otsu thresholding with spatial 2D filtering to enhance character recognition in low-quality or handwritten documents.
3. A contextual NLP pipeline combining structured pattern detection, keyword-context verification, and Named Entity Recognition (NER) to classify document risk levels, reducing reliance on isolated static keyword matching alone.
4. A deep learning classifier employing EfficientNetV2-S capable of performing five-class forensic identification across different covert hiding methods.
5. A decision-making framework that combines visual cues, semantic meaning, and forensic evidence into one secure enterprise policy.

II. PROBLEM DOMAIN

This research focuses on the overlap between Information Security specifically Data Loss Prevention and Covert Channel Detection and Multi-Modal Artificial Intelligence involving

Computer Vision and Natural Language Processing. The specific areas covered include:

- **Heterogeneous Document Processing:** Dealing with diverse enterprise and public-sector documents marked by significant variations within classes such as low-resolution mobile pictures, rotated scans, deteriorating paper quality, handwritten notes from clinical settings, and official seals in different languages.
- **Covert Digital Communications:** Hidden data channels in digital image transfers where confidential information is concealed within either the spatial arrangement or frequency analysis of cover images without changing how the carrier image appears.
- **Regulatory Compliance Boundaries:** Automatically enforcing data privacy laws like GDPR and HIPAA at points where automated communications exit systems as well as during uploads to internal databases.

III. PROBLEM DEFINITION

Current digital egress inspection systems struggle with unstructured documents because they only consider data exposure through clear, machine-readable text. This leads to two major issues that weaken enterprise Data Loss Prevention (DLP):

- **Semantic and Perceptual Vulnerability (Overt Leakage):** Standard rule-based inspection tools and keyword lists fail to understand the organizational context of unstructured image data. When documents have visual distortions, variations in handwriting, rotated layouts, or intricate formatting (like medical forms, handwritten prescriptions, police reports, or government IDs), character recognition processes perform poorly. This allows sensitive information to slip past security measures unnoticed.
- **Structural Blindness to Covert Channels (Steganographic Exfiltration):** Traditional DLP systems only check the visible text and images on the surface of a document. Malicious users take advantage of this by hiding stolen data within pixel patterns or using steganographic techniques like Least Significant Bit (LSB), Pixel Value Differencing (PVD), Discrete Cosine Transform (DCT), and Fast Fourier Transform (FFT). Since the modified image looks perfectly normal to human reviewers and OCR software, these hidden extractions escape detection without raising any security alarms.

For any random digital image sent across a network boundary, this system aims to create an automated decision-making framework that assesses the file based on two different security standards:

- **Visible Privacy Violation Analysis:** The system should analyze the raw image to identify visual layout features and embedded text tokens. It will categorize these tokens according to their high-level meaning and check if the document contains protected information such as bank statements, legal reports about crimes, patient medical records, or official identification documents—all without relying on specific training for each task or fragile regex patterns.
- **Covert Artifact Detection:** At the same time, the system must examine pixel-level data and frequency statistics to determine if the image is genuine or has hidden content inside it. If any steganographic manipulation is found, the system must classify it into one of five categories: Clean, Least Significant Bit (LSB), Pixel Value Differencing (PVD), Discrete Cosine Transform (DCT), or Fast Fourier Transform (FFT).

To create a strong defense against security threats, the solution must meet these requirements:

- I. **Fail-Secure Decision Logic:** If any part of the visual layout analysis, semantic text classification, or forensic steganalysis indicates a high-risk issue, the system must trigger an alert and restrict access to the document.
- II. **Zero-Shot Adaptability:** The semantic intent classifier needs to identify sensitive business categories using natural language understanding instead of fixed regex patterns so that it can adapt to new document formats easily.
- III. **Forensic Granularity:** The steganalysis process should not just serve as a simple yes-or-no detector; it should pinpoint how data was embedded in order to give useful forensic information for security teams in enterprises.

IV. LITERATURE REVIEW

In recent years, the use of images to document, share and store identification cards, screenshots and other information has raised new questions regarding data privacy and leakage prevention. In contrast to traditional Data Loss Prevention (DLP) solutions, which focus mainly on structured or directly accessible text information, image-based information can carry sensitive information in visual text and information that is hidden in image data. In recent years, computer vision, Optical Character Recognition (OCR), Natural Language Processing (NLP), and machine learning methods have been investigated to detect and safeguard sensitive data within images.

Recent techniques have been catching more and more PII (Personally Identifiable Information) automatically. REDACT suggested an automated framework for redaction of PII using a combination of document processing and an AI based analysis phase to detect sensitive information and prevent it from being disclosed [1]. Likewise, recent studies

on visual privacy have shown that much of the information that is present in natural images can constitute a serious privacy risk in itself, without the need for any modifications. To overcome this issue, TextSafety was developed to identify text regions in the image and reconstruct the surrounding area to remove the text, leaving no visible text in the restored image [2]. The methods presented here illustrate the need to recognize and preserve textual information within images and not image as an undifferentiated visual entity.

Therefore, OCR has become a significant part of the extraction of information from image data and protection of privacy. A new secure document processing system leveraged OCR and locally deployed LLMs to extract text from scanned documents, and used semantic and contextual analysis to ensure data privacy [3]. Semantic analysis following OCR is especially relevant since the presence of text does not necessarily mean that there is information leakage. The meaning of a text sequence, its context, and its sensitivity. This encourages the use of Named Entity Recognition (NER) and contextual analysis following OCR to separate the potentially sensitive entity from the normal text. In the proposed scheme, OCR also provides the information in textual form along with word level (x, y) coordinates and confidence scores for further NLP based sensitivity analysis.

The protection of privacy has also made a step from just marking images as privately sensitive to selectively obfuscating sensitive areas and localizing sensitive areas. PrivObfNet introduced a weakly supervised semantic segmentation model which classifies attributes related to privacy and automatically generates privacy-sensitive region obfuscation regions [6]. This method shows that privacy protection is not only possible by providing a single privacy label to an image, but that spatial localization can also provide privacy protection benefits. Another approach focused on the protection of private information in tweets, images, memes, and metadata, specifically on information leakage from text that is embedded in visual media [7]. Such studies confirm the need to maintain the usability of an image while protecting only the areas that contain sensitive data. The proposed system is based on this; it can identify sensitive content in a text and highlight or redact the part of the image where the sensitive content is detected.

While visible text is one leakage path, sensitive information can be embedded in the pixels of an image without revealing any text. The field of steganalysis research holds therefore another very important path in the development of image security. The recent research efforts have focused on steganalysis methods using deep learning that can detect stealthy statistical and visual signatures of the hidden information embedded in images. A recent deep-learning based steganalytic method for generative steganography was proposed in [4] where the need for more specialized detection methods was also emphasized, since traditional steganalysis methods might not be suitable to detect generative steganography. In [4] the necessity for more specialized detection methods was also pointed out since the traditional steganalysis methods might not be effective for

newer generative steganography techniques, and a recent deep-learning based steganalytic method for generative steganography was proposed. Likewise, GIFDL explored the deep learning approach to distortion modeling for enhancing the security aspects of image steganography which further complicated the generation process and the embedding method with respect to steganalysis [5].

Other recent work conducted in the IEEE has also focused on steganalysis as a way of detecting hidden messages in images. Steganalysis on Dual-Layer Security of Messages Using Steganography and Cryptography was about the detection of information steganographically placed in a security context [8]. These are all signs that hidden information can only be achieved by analysing the image characteristics which is not possible through the conventional OCR approach. The proposed implementation classifies images into clean and multiple steganographic images (LSB, PVD, DCT, FFT) using the EfficientNetV2-S, and a chi-square statistical test for additional evidence of LSB-based hidden images.

In sum, the current literature shows that a great deal of work has been done in automated detection of PII, visual-text privacy protection, selective image obfuscation, and steganalysis using deep learning. But in general these research directions focus on problems of visible sensitive information and problems of hidden information separately. The main difference between the OCR-based approach and steganalysis approach is that the former concentrates on extracting and classifying visible texts, whereas the latter concentrates on detecting information hidden inside the image's pixels. Also, image privacy approaches generally focus on either visual features or text extraction, but not on both. Likewise, visual features or text detection approaches are the ones that are generally used in image privacy rather than dealing with semantic sensitivity analysis and hidden-information detection. This is a chance to develop an integrated image-based DLP solution that can take into account several leakage ways.

This gap is addressed by proposing an image security system, which integrates OCR-based text extraction, NER and contextual sensitivity analysis, vision-based supporting evidence, and deep-learning-based steganalysis. The results from these components are combined and confidence scores and risk categories are assigned and range from REVIEW to CRITICAL. Sensitive areas detected via OCR are then highlighted or redacted, and secure reports and audit information are produced while not retaining unneeded sensitive information. This integrated approach is the extension of current image privacy and steganalysis solutions towards a more complete image-based Data Loss Prevention (DLP) system for uncovering visible and hidden sensitive information.

V. METHODOLOGY

The proposed system is an image-based Data Loss Prevention (DLP) framework designed to detect potential leakage of sensitive information from unstructured image

content. The methodology addresses visible sensitive information through OCR, pattern-based detection, contextual NLP, Named Entity Recognition (NER), and image-level visual analysis, and is supported by a steganalysis module that examines whether information may be concealed within the image data.

A. Overall System Architecture

The framework accepts an image as input and performs image enhancement followed by complementary text-based and image-based analysis. The visible-information pathway identifies sensitive information explicitly present in the image, while the steganalysis pathway examines whether information may be concealed within the image pixels.

For visible-information analysis, the input image is first subjected to adaptive preprocessing to improve the quality of subsequent OCR. Tesseract OCR then extracts textual content along with word-level confidence values and spatial coordinates. The extracted text is analyzed using pattern-based sensitive-information detection and contextual NLP/NER, with document-level sensitivity indicators considered as supporting evidence.

The original image is also classified using a lightweight OpenCV structural-heuristics classifier. When the CLIP (ViT-B/32) backend is available, it is run as well, and its prediction replaces the OpenCV result whenever CLIP reports equal or higher confidence. The vision branch is therefore a complement to the text-based analysis rather than an alternative to it, and remains available even when CLIP is not installed.

A shared confidence and risk-assessment engine evaluates the outputs of these stages. Detected sensitive regions can then be highlighted or redacted using the spatial information obtained during OCR, after which the system generates a structured, encrypted report with audit information.

The steganalysis module operates as an additional security layer, using EfficientNetV2-S to distinguish clean images from images containing supported steganographic techniques (LSB, PVD, DCT, and FFT).

The overall pipeline is summarized as:

Input Image → Preprocessing → OCR → Sensitive Information Analysis → Vision Analysis → Evidence/Risk Assessment → Annotation/Redaction → Secure Reporting

with steganalysis providing an additional, parallel covert-information analysis pathway.

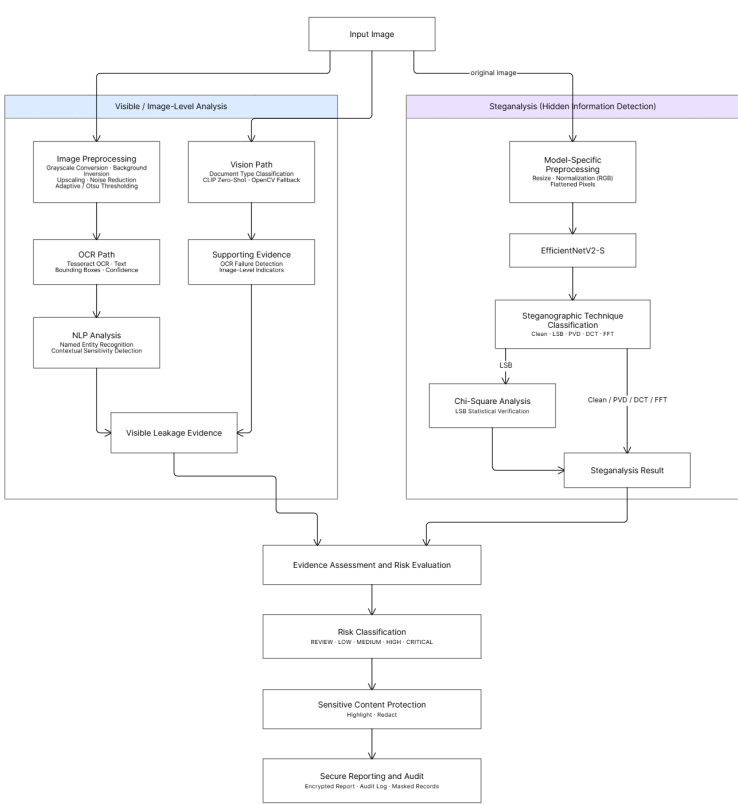


Fig. 1. System architecture of the proposed dual-pathway image DLP framework.

B. Image Preprocessing

Image preprocessing improves the reliability of OCR across heterogeneous image conditions, covering scanned documents, photographs, screenshots, and identification documents that differ substantially in resolution, illumination, background intensity, and noise. The pipeline applies image-dependent operations rather than a single fixed transformation.

The input image is first converted to grayscale. Mean pixel brightness is then checked, and images with a mean brightness below 127 (predominantly dark backgrounds) are automatically inverted to improve foreground–background separation. The minimum image dimension is then examined; images whose smaller side is below 1000 pixels are upscaled by a factor of 2 using bicubic interpolation, and the applied scale factor (1.0 or 2.0) is retained for later coordinate remapping.

Noise is then reduced using Fast Non-Local Means denoising (h = 10, 7×7 template window, 21×21 search window). To handle illumination variation, the image is divided into a 4×4 grid, and the variance of block-wise mean intensity is computed. If this variance exceeds 500, Gaussian adaptive thresholding is applied (31×31 neighborhood, C = 10); otherwise, Otsu's thresholding is used.

The preprocessing stage produces a binarized representation for OCR while retaining the scale factor introduced during

resizing, allowing OCR-derived spatial coordinates to be mapped back to the original image dimensions — necessary for sensitive-region localization and redaction.

C. OCR-Based Visible Information Extraction

The preprocessed image is analyzed using the Tesseract OCR engine (via `pytesseract`) to obtain machine-readable text. The OCR stage retains recognized text along with word-level confidence values and bounding-box coordinates.

The implementation supports three Page Segmentation Modes: PSM 3 (fully automatic layout detection, used as a general fallback), PSM 6 (assumes a single uniform block of text; used as the default, since it performs best for scans and screenshots), and PSM 11 (sparse-text mode, used for ID cards and forms where text is scattered).

Words with a Tesseract confidence score below 60 (on a 0–100 scale) are discarded, along with blank or whitespace-only rows, before the remaining words are joined into the text string used by downstream analysis. The stage also records the raw word count, filtered word count, and resulting dropped-word ratio, which is later used as a quality signal by the confidence engine. OCR is thus not treated merely as a text-extraction step — it also supplies the spatial and quality evidence used by the subsequent DLP stages.

D. NLP-Based Sensitive Information Detection

The extracted OCR text is analyzed using complementary pattern-based and contextual NLP techniques. The pattern-based stage detects structured identifiers, while contextual analysis and NER provide additional semantic evidence.

The implemented DLP detector supports fourteen sensitive-information categories: email addresses, Indian mobile numbers, PAN, passport numbers, GST numbers, IFSC codes, SWIFT/BIC codes, Aadhaar numbers, bank/card numbers, voter IDs, MICR codes, dates of birth, driving licences, and MRZ lines. Each pattern type carries a base confidence tier reflecting how unambiguous it is: PAN, passport, GST, IFSC, SWIFT and email are high-confidence patterns with near-zero false-positive risk; Aadhaar, bank/card numbers, voter ID and MICR are medium-confidence; and date-of-birth and driving-licence numbers are low-confidence, context-dependent patterns.

For each candidate match, a window of 8 words on either side is checked against a sensitive-context keyword set and a benign-context keyword set. A benign-context overlap takes precedence and flags the finding as false-positive risk regardless of the pattern's own confidence tier; in the absence of benign context, an overlap with the sensitive-context set upgrades the finding's confidence tier instead. This directly resolves the ambiguity of values such as dates and generic numeric sequences that can occur in both sensitive and non-sensitive contexts.

The NLP stage also performs Named Entity Recognition to identify entities such as persons and organizations not captured by structured patterns, using a lightweight spaCy-based NER model with a Hugging Face BERT-based NER model available as an alternate backend. Detected entities can subsequently be mapped to relevant DLP categories. Document-level sensitivity indicators (e.g., "confidential" or "restricted" labels) are considered as additional evidence. The resulting NLP analysis therefore combines structured identifiers, contextual verification, named entities, and document-level indicators rather than relying on isolated keyword or regex matches alone.

E. Vision-Based Image Analysis

The vision-based component supplies image-level evidence that complements the OCR and NLP analysis, primarily to reduce dependence on successful text extraction for structured images where OCR may produce insufficient or unreliable output.

An OpenCV-based structural classifier is always computed first, using image statistics such as edge density, rectangular-region count, and text-like region count to estimate a document type — `id_card`, `passport`, `cheque`, `document`, `screenshot`, or `unknown` — with a heuristic confidence score. When the CLIP (ViT-B/32) backend is available, it is loaded on CPU and run as a zero-shot classifier over the same categories; if CLIP's predicted type is not "unknown" and its confidence is greater than or equal to the OpenCV estimate, CLIP's prediction is used instead. This keeps the vision stage fully operational even when CLIP or its dependencies (torch/timm) are unavailable, while letting CLIP improve accuracy when present.

Each document type maps to a fixed sensitivity level (`cheque` → `CRITICAL`; `id_card` and `passport` → `HIGH`; `document` and `screenshot` → `MEDIUM`; `unknown` → `LOW`), and confidence is bucketed as high (≥ 0.75), medium (≥ 0.45), or low (< 0.45). A further objective of this branch is identifying potential OCR-failure risk: an image may exhibit document-like structural characteristics while yielding very few recognized words, and this is flagged separately so that low OCR output is not mistaken for the image containing no sensitive content. Vision output is treated as supporting rather than determinative evidence within the overall risk assessment.

F. Steganalysis-Based Detection of Hidden Information

A separate leakage pathway considered by the framework is the concealment of information within image data through steganographic techniques, which may not be visually apparent and therefore cannot be reliably detected through OCR or visible-content analysis.

The module formulates steganalysis as a five-class image-classification task: `Clean`, `LSB`, `PVD`, `DCT`, and `FFT`. An EfficientNetV2-S convolutional neural network with ImageNet-pretrained weights serves as the classification model, trained on a combination of clean landscape images

and datasets representing each supported steganographic technique. Before inference, images are converted to RGB and transformed via padding, center cropping/resizing to 256×256 pixels, and ImageNet-based normalization; the classification head is adapted to the five target classes.

Training initially freezes the pretrained backbone and trains only the task-specific classification layer using class-weighted cross-entropy loss, followed by progressive fine-tuning of the backbone at a lower learning rate. The model is evaluated using validation accuracy and class-wise F1 scores. In addition to the learned classifier, chi-square statistical analysis provides supplementary evidence for LSB-based concealment.

This module is intentionally kept separate from the primary visible-information pathway: OCR/NLP/vision analysis addresses information accessible from the image's visible content, while steganalysis investigates information potentially concealed within the image representation itself.

G. Evidence Fusion and Risk Assessment

Findings from the pattern, contextual/NLP, and OCR-quality stages are combined by a unified confidence-scoring engine into a single score in the range [0, 1]:

$$\text{unified_score} = 0.50 \times \text{regex_score} + 0.35 \times \text{nlp_score} + 0.15 \times \text{ocr_score}$$

The regex component reflects pattern-confidence tier (high = 1.00, medium = 0.65, low = 0.35, none = 0.00 for NLP-only findings); the NLP component reflects context-verification/NER outcome (high = 1.00, medium = 0.65, low = 0.30, spaCy default = 0.65, neutral 0.50 when no NLP signal is available); and the OCR component reflects overall scan quality — good (mean confidence ≥ 70 , full weight), moderate (≥ 50 , partial weight), or poor (< 50 , reduced weight) — with an additional penalty applied when more than half of all OCR words were dropped.

The resulting score maps to one of five risk categories: CRITICAL (≥ 0.85), HIGH (≥ 0.70), MEDIUM (≥ 0.50), LOW (≥ 0.30), and REVIEW (< 0.30 , or forced whenever benign context flags a finding as false-positive risk, regardless of its numeric score). REVIEW is thus reserved for findings where the available evidence is insufficient — or actively contradictory — to treat the detection as confirmed leakage. An image's overall risk is the highest non-REVIEW risk level among its findings, falling back to REVIEW only if every finding required review.

Vision-based evidence is not incorporated as a direct input to the unified score; instead, it supports the assessment indirectly — most notably through the OCR-failure-risk flag, which prevents an image with genuinely little text from being treated the same as one where limited OCR output reflects a vision-detected document structure. The steganalysis result adds an independent security dimension, so the final assessment reflects both visible

sensitive-information exposure and potential covert concealment.

H. Sensitive Region Annotation and Redaction

The framework spatially localizes detected sensitive information using the word-level bounding boxes obtained during OCR. When a finding is identified, its constituent OCR word boxes determine its location within the original image; where upscaling was applied during preprocessing, these coordinates are divided by the retained scale factor to remap them onto the original image dimensions, keeping the detected region correctly aligned with the source image.

Two output modes are supported: highlighting, for analyst verification and visual inspection, and redaction, for obscuring the detected region and producing a privacy-preserving image. The framework therefore does not stop at a document-level sensitivity label — it also localizes detected sensitive content, enabling selective protection of the affected regions.

I. Secure Reporting and Audit Management

Following analysis, the framework generates a structured report containing detection results, masked and full values, and associated metadata. By default, this report is encrypted using Fernet symmetric encryption (AES-128-CBC with HMAC-SHA256 authentication) from the Python `cryptography` library, with the key generated on first run and stored separately from the report; a plain-JSON mode is also available for debugging but is not the default. A separate audit log records finding-type counts and timestamps only, without sensitive values, so it can be safely stored unencrypted. The system also maintains masked training records in which sensitive values are sanitized before storage.

The final output therefore consists of the DLP assessment, detection evidence, optional highlighted or redacted images, an encrypted report, and audit information.

VI. RESULTS

The proposed Image Data Loss Prevention (DLP) framework was tested with representative image-based data sources which might be sources of sensitive information. The visible-information evaluation focused on identity documents, enterprise-related screenshots, such as driving licenses, passports, ID cards, internal system screenshots, and banking information screenshots. The type of sensitive information was detected, confidence of detection, risk level assignment, number of findings and the ability to localize sensitive regions in the image were all taken into account when evaluating.

The second channel of the framework: steganalysis based on EfficientNetV2-S, was also evaluated. The steganalysis module is capable of identifying five different classes: “Clean”, “Least Significant Bit (LSB)”, “Pixel Value Differencing (PVD)”, “Discrete Cosine Transform (DCT)”,

and “Fast Fourier Transform (FFT)”. A statistical test (chi-square) is also included for LSB-based analysis.

a.) Sample Input Images

The images in Figure 1 and Figure 2 are representative identity-document images that demonstrate the visible sensitive-information detection task. Personal identifiers, document numbers, dates, nationality or address related information and other attributes of identity are included in the documents. The following illustrations show the type of unstructured visual content that needs OCR and contextual analysis.



Fig. 2. Sample driving-license image used for visible-information evaluation.



Fig. 3. Sample passport image used for visible-information evaluation.

B. Sensitive Information Detection

The visible-information pipeline includes OCR, entity recognition, context analysis, and document-level sensitivity indicators for detecting potentially sensitive information. The categories of sensitive information relevant to the analyzed image types are summarized in Table I. The values of confidence and risk reported are current values.

Image Category	Sensitive Information Detected / Relevant Fields	Confidence	Risk Level
Internal System Screenshot	Employee name, employee ID, email address, username, internal IP/host information,	0.67	Critical

	organization/confidential labels		
Driving License	License number, person name, date of birth, address, issue/expiry date	0.90	Critical
Passport	Passport number, person name, date of birth, nationality, sex, expiry date, MRZ information	0.88	Critical
ID Card	Person name, ID/enrollment number, date of birth, address, identity information	0.79	High / Sensitive
Banking Information Screenshot	Account number, account-holder name, IFSC/bank identifier, transaction information, phone number, card/payment information	0.67	Critical

TABLE I
SENSITIVE-INFORMATION CATEGORIES AND REPORTED CONFIDENCE/RISK VALUES

In the case of driving license information, the information that is sensitive is the license number, person name, date of birth, address, issue or expiry data. The passport example includes similar identity attributes along with information relating to passports and MRZ. There can be multiple heterogeneous sensitive entities in banking and internal system screenshots, which is why it is crucial to perform a contextual analysis, as not all text strings in a screenshot constitute a privacy breach.

C. Confidence Score Analysis

The range of confidence scores reported in the five categories of images considered here is from 0.67 to 0.90, and the average level of confidence is 0.782. The driving-license case reported the highest level of confidence (0.90), followed by the passport (0.88) and ID-card (0.79) cases. The internal-system and banking screenshots resulted in confidence values of 0.67. The variation suggests that the layout of the image, quality of text, and heterogeneity of content may impact the evidence that is available to the OCR and contextual-analysis stages.

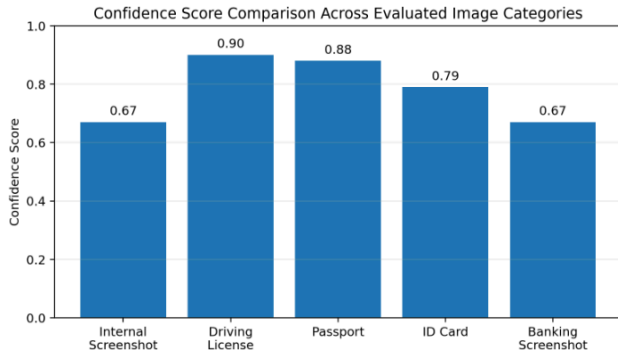


Fig. 4. Confidence score comparison across the evaluated image categories.

D. Risk-Level Distribution

The evidence-fusion stage converts the detected information and supporting signals into risk categories. For the 5 reported evaluation cases, 4 of them were given Critical risk and 1 case was given a High / Sensitive risk. This equates to 80% Critical and 20% High / Sensitive cases.

Risk Level	Number of Test Cases	Percentage
Critical	4	80%
High / Sensitive	1	20%
Total	5	100%

TABLE II
RISK-LEVEL DISTRIBUTION FOR THE REPORTED EVALUATION CASES

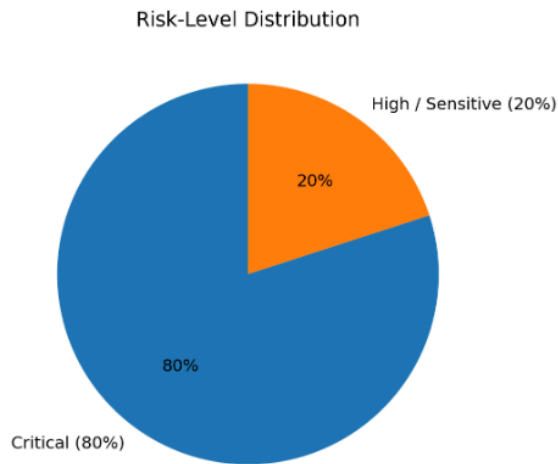


Fig. 5. Risk-level distribution across the evaluated cases.

E. Comparison of Reported Sensitive Findings

There is variability in the number of sensitive findings reported by image type. The internal system screenshot had the most number of findings reported (10), then the banking information screenshot (4), driving license (3), and passport (2). No separate finding count was recorded for the

evaluation in the ID-card case. The variation illustrates the ability of the framework to output not one, but several sensitive regions for a single image, not just a document-level label of "sensitive" or "not-sensitive".

Image Category	Reported Findings
Internal System Screenshot	10
Banking Information Screenshot	4
Driving License	3
Passport	2
ID Card	Not recorded
Total (recorded)	19

TABLE III
COMPARISON OF REPORTED SENSITIVE FINDINGS ACROSS IMAGE CATEGORIES

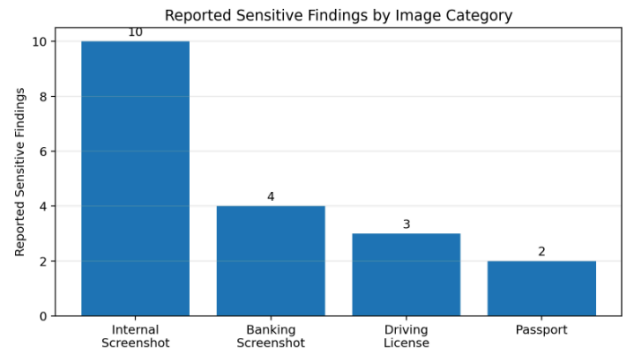


Fig. 6. Reported sensitive findings by image category.

F. Steganalysis Evaluation

The second evaluation channel looks at information that could be embedded in the image's pixels, and thus not be reliably identified solely with visual inspection or OCR. The EfficientNetV2-S classifier is trained as a multi-class (5 classes) classification model for the Clean, LSB, PVD, DCT and FFT images. The accuracy of the five-class steganalysis task is reported to be 78.40%. In addition, a chi-square statistical test is used as supplementary evidence in support of concealment by LSB.

Evaluation Component	Configuration	Reported Result
Steganalysis classifier	EfficientNetV2-S	78.40% validation accuracy
Classification task	Clean, LSB, PVD, DCT, FFT	5 classes
Additional statistical analysis	Chi-square test	LSB-based analysis
Model input	256 × 256 RGB representation	As specified in methodology

TABLE IV
STEGANALYSIS EVALUATION CONFIGURATION AND REPORTED RESULT

Class	Technique	F1 Score
Clean	Original, non-stego images	0.985
LSB	Least Significant Bit	0.866
PVD	Pixel Value Differencing	0.557
DCT	Discrete Cosine Transform-based	0.435
FFT	Frequency-transform-based	0.862

TABLE V

CLASS-WISE F1 SCORES FOR THE FIVE-CLASS STEGANALYSIS TASK

Beyond the aggregate validation accuracy of 78.40%, class-wise F1 scores show substantial variation across the five steganographic techniques. The model performs strongly on Clean (F1 = 0.985), LSB (F1 = 0.866), and FFT (F1 = 0.862) images, indicating that these categories leave more detectable statistical or pixel-level artifacts. Performance is markedly lower for PVD (F1 = 0.557) and DCT (F1 = 0.435), reflecting the subtler, more adaptive nature of these embedding techniques, which introduce smaller and less uniform distortions that are harder for the classifier to distinguish from clean images. This class-wise breakdown indicates that the aggregate accuracy figure alone understates the difficulty of detecting certain steganographic techniques, and that future work should specifically target improved detection of PVD- and DCT-based concealment.

G. Comparative Evaluation of Detection Capabilities

A capability-level comparison shows that the traditional workflow typically used for OCR and the proposed workflow for integration are different in their functionality. OCR will be able to read visible text, but it isn't the only means of identifying information hidden in the pixels of a picture. The proposed approach integrates visible information analysis, a dedicated steganalysis channel and evidence fusion stage.

Detection Capability	OCR/NLP-Oriented Pipeline	Proposed Integrated Framework
Visible text extraction	Yes	Yes
Contextual sensitive-information analysis	Partial	Yes
Document-level sensitivity analysis	Limited	Yes
Spatial localization of findings	Yes	Yes
Risk classification	Limited	Yes
Hidden-information detection	No	Yes

LSB/PVD/DCT/FFT classification	No	Yes
LSB statistical analysis	No	Yes
Highlighting/redaction support	Not inherent	Yes

TABLE VI

CAPABILITY-LEVEL COMPARISON OF VISIBLE-INFORMATION AND INTEGRATED IMAGE-DLP ANALYSIS

H. Annotated Detection Output

The word-level spatial coordinates obtained in the OCR process are used to annotate the original image during the annotation stage, thereby locating sensitive regions in the original image. If the image was upscaled, the coordinates are remapped to the original image dimensions in the preprocessing. The framework enables highlighting for analyst verification and redaction for privacy-preserving output.



Fig. 7. Sensitive-region detection for the supplied driving-license image.

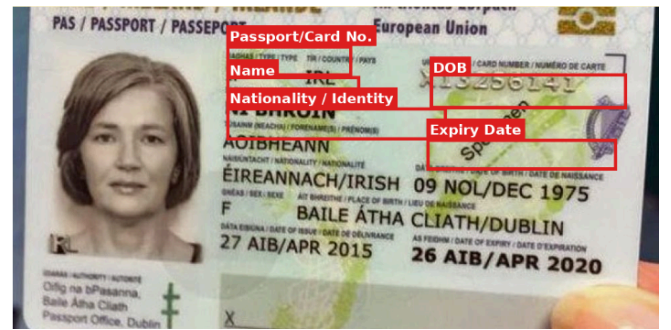


Fig. 8. Sensitive-region detection for the supplied passport image.

I. Overall Evaluation

The evaluation shows the desired behavior of the proposed framework in two channels. The system provides document-sensitive results for the visible-information channel as well as confidence and risk information. The range of reported confidence values is from 0.67 to 0.90 and the risk distribution has four Critical cases and one High /

Sensitive case. Multiple sensitive entities can also be reported from a single image in the finding-count analysis.

The second channel adds analysis of content that is not visible on the page. EfficientNetV2-S offers five-class steganalysis for Clean, LSB, PVD, DCT and FFT images with the reported validation accuracy of 78.40%. The added chi-square analysis gives statistical evidence for the concealment based on LSB. These components aim to tackle two aspects of image-leakage: visible information and potential hidden information in image data.

The results should be interpreted in terms of the evaluation available, however. The visible-information experiment has been conducted on a small sample of representative categories of images, and the results (in terms of the confidence and risk percentages) are not to be taken as representative of the performance of the general population. A more comprehensive assessment would integrate more ground-truth annotated data, class-wise precision, recall, F1-score, confusion matrices, as well as controlled comparisons with baseline approaches. Handwritten, low resolution, distorted and poorly lit images may also cause a reduction in OCR performance. These restrictions offer clear guidelines for further testing and development.

VII. CONCLUSION & FUTURE WORK

This paper introduces a unified Data Loss Prevention (DLP) framework based on two main pillars. It combines multi-modal contextual intelligence with deep learning techniques for image steganalysis to identify both clear compliance violations and hidden data theft attempts in unstructured digital content. The system uses OCR combined with pattern-based detection, contextual keyword verification, and NER to identify sensitive text, an adaptive OpenCV preprocessing pipeline to improve character recovery, and an OpenCV/CLIP-based structural classifier to support document-type identification. This allows it to accurately identify high-risk documents—like financial records, police FIRs, medical prescriptions, and biometric IDs—without relying on fragile pattern-matching rules. Additionally, the EfficientNetV2-S forensic module can detect hidden data using spatial (LSB, PVD) and frequency-domain (DCT, FFT) steganography techniques with a validation accuracy of 78.40%. The resulting decision-fusion policy creates a strong security barrier, while future work will focus on improving detection

accuracy for adaptive embedding methods (PVD and DCT), implementing real-time automated redaction processes, and reducing inference delays for large-scale business applications.

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